Pathophysiology of edemas, infusion therapy and cerebral edema

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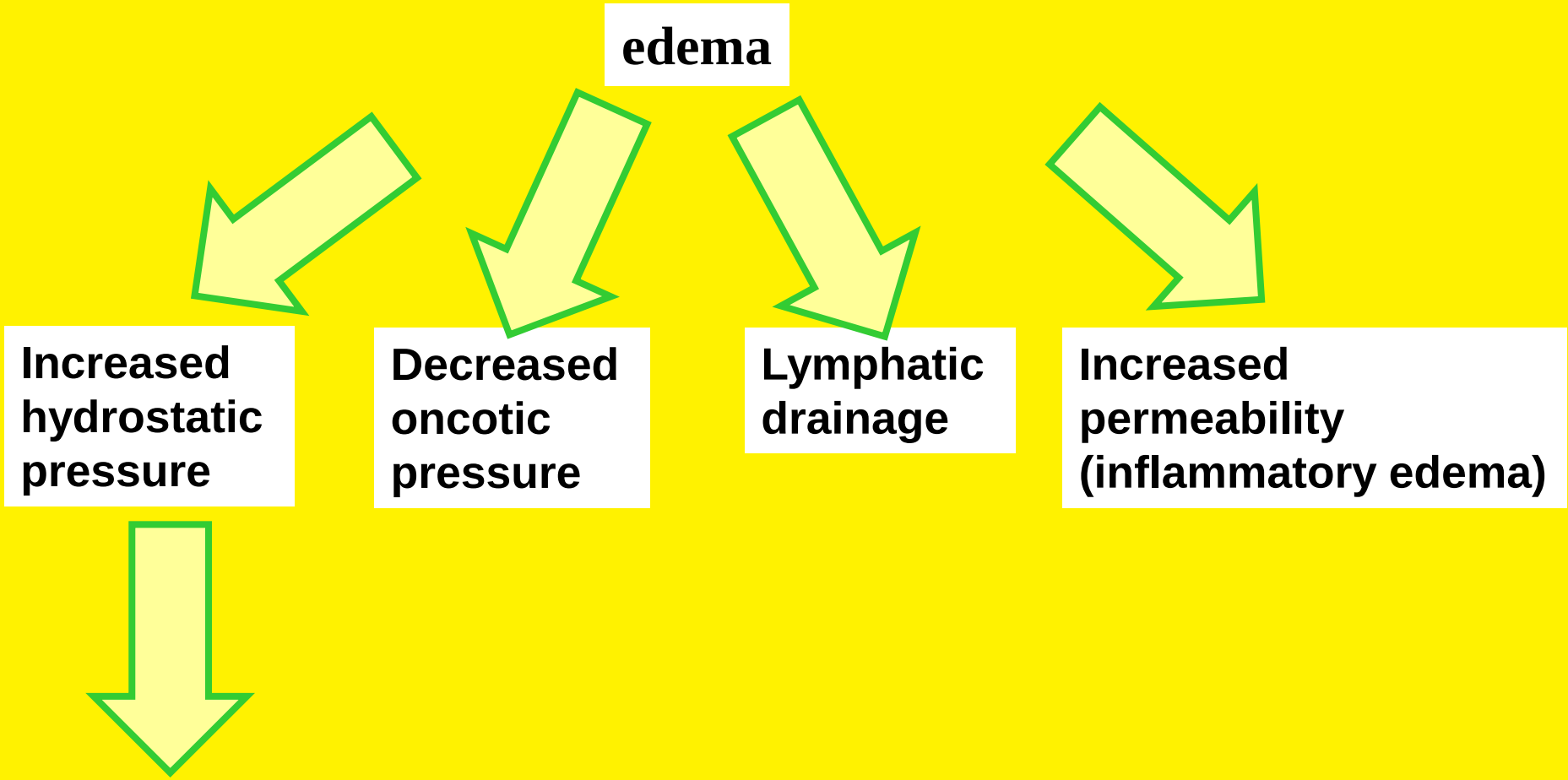
Pathogenesis of swelling

- Edema:

  1. Interstitial edema is a disturbance of starling capillary forces = Capillary $\Leftrightarrow$ interstitium exchange (note: It is not same as Frank–Starling law). This presentation is primarily about interstitial edema, shortly edema.

  2. Cellular edema: swollen or shrinkage of red cells in the states of hyperosmolality or hypoosmolality (mainly: Na^{+})

1. Edema: classification based on cause



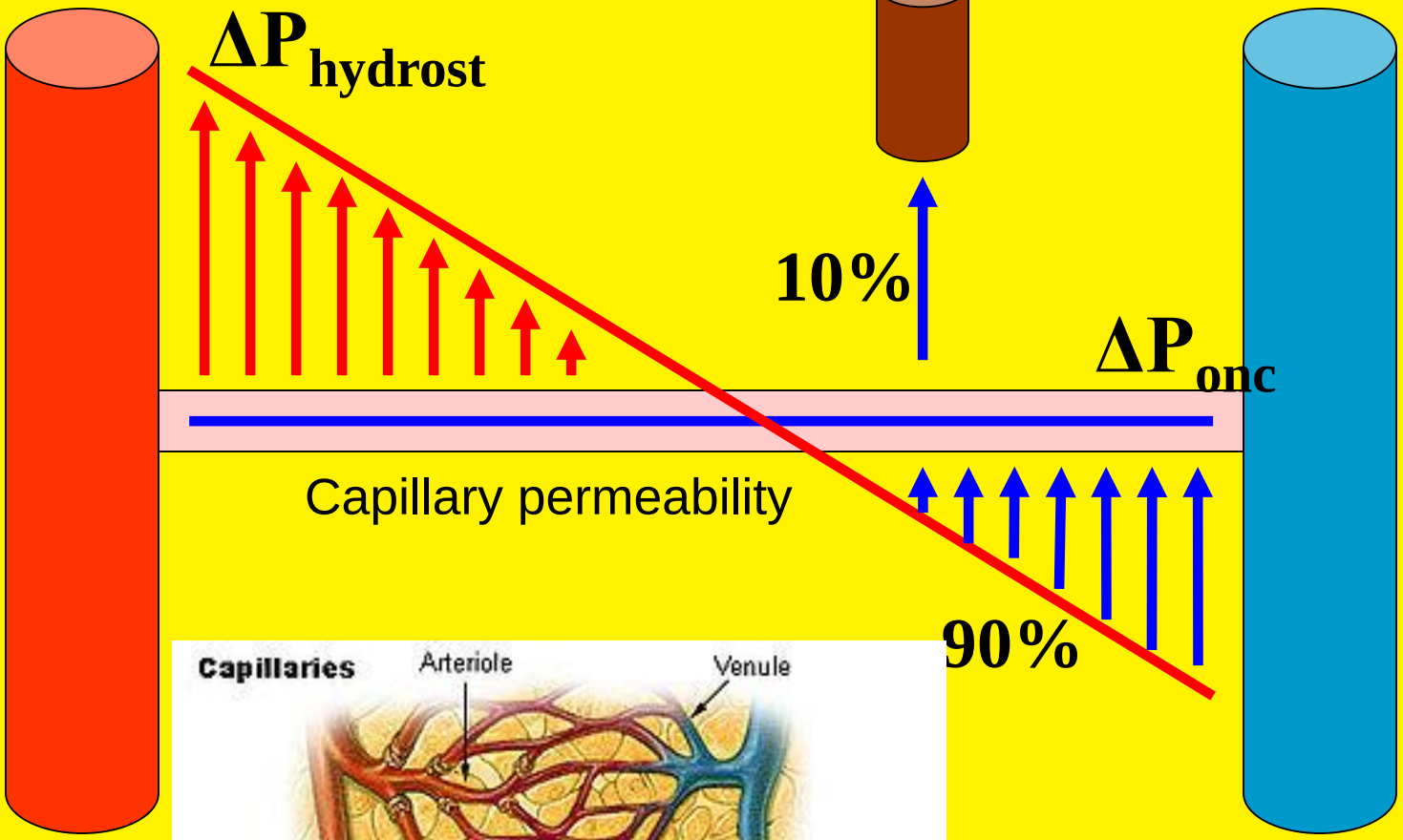
A: Venous pressure increase
B: Kidney water and salt retention
C: Arteriolar dilation

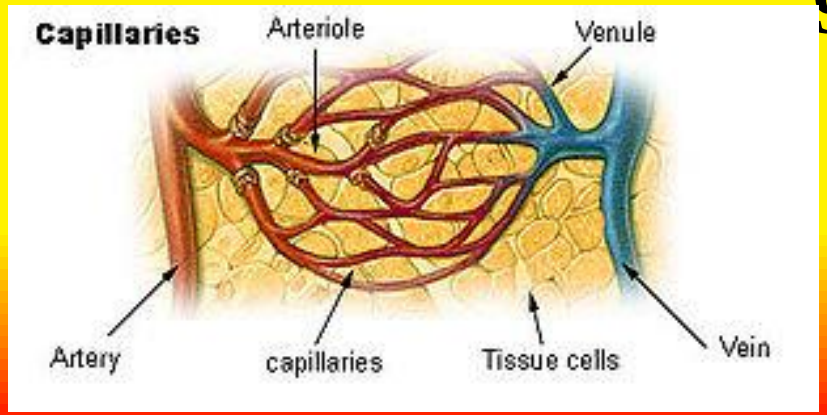
Basics: Starling capillary forces
Capillary ⇔ interstitium exchange

Arterial end

Venous end

Lymphatic drainage





1. Edema: classification based on cause

Increased hydrostatic pressure A

A: Venous pressure increase

Generalized:

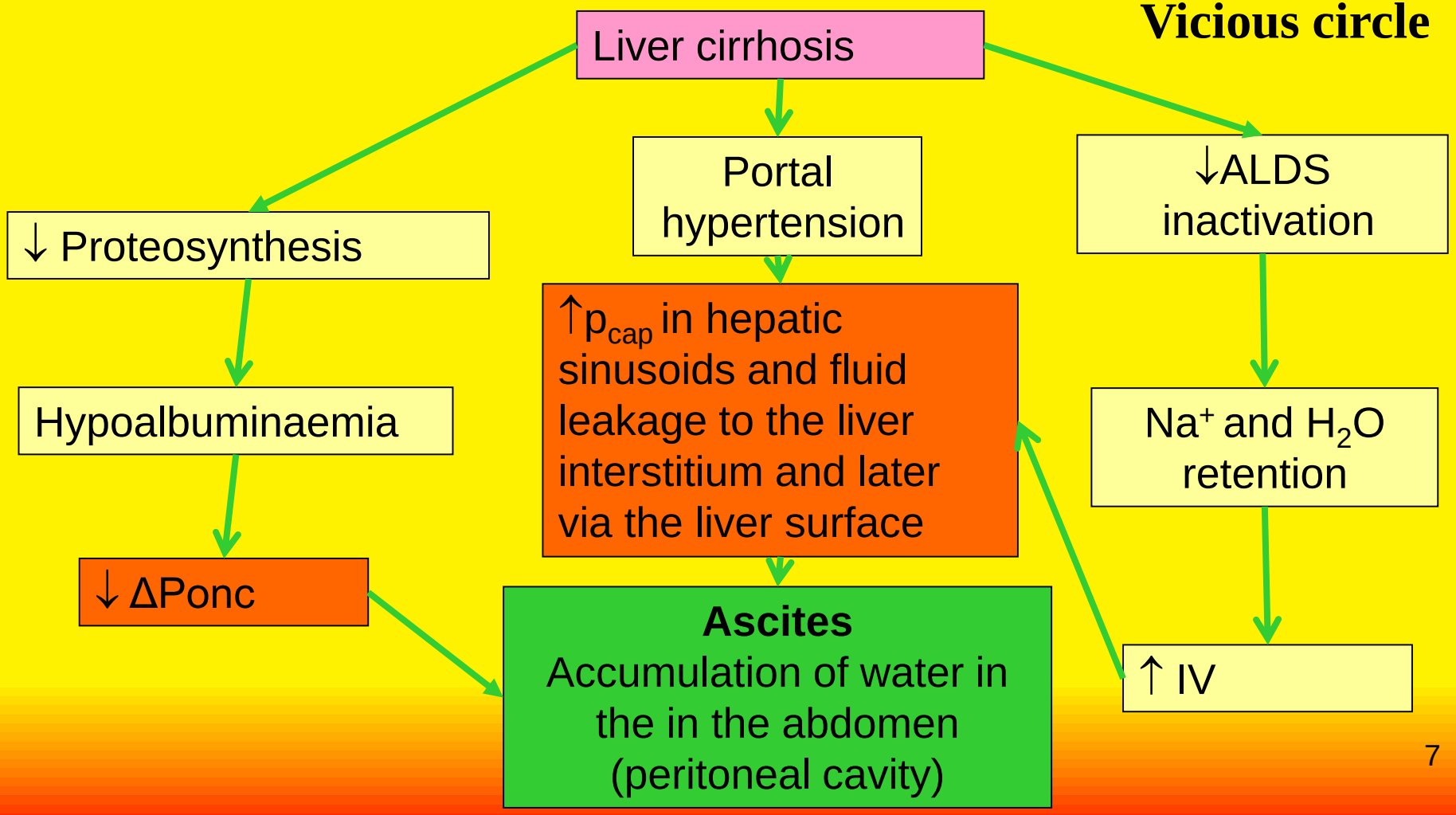
- Right congestive heart failure: venous obstruction causes the increased hydrostatic pressure behind the obstruction and swelling in the legs, ankles, and liver
- Cirrhosis of the liver

Local:

- Venous insufficiency
  - Phlebothrombosis = deep vein thrombosis (or the presence of a clot within a vein, unassociated with inflammation due to Virchow's triad) may lead to pulmonary embolism
  - Superficial thrombophlebitis: phlebitis = inflammation of a superficial vein (e.g. skin infection or varicose veins) has preceded the formation of a thrombus and may lead to varicose veins
- Tumor oppression

Ascites: liver cirrhosis

- Accumulation of water in the in the abdomen (peritoneal cavity)
- the most common cause is liver cirrhosis



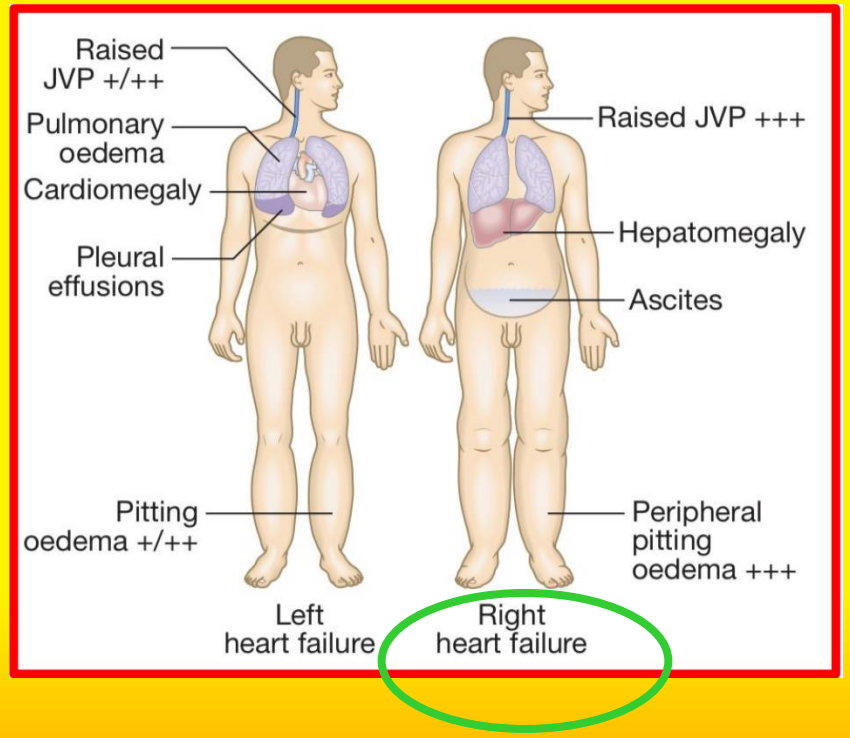
Ascites: other causes

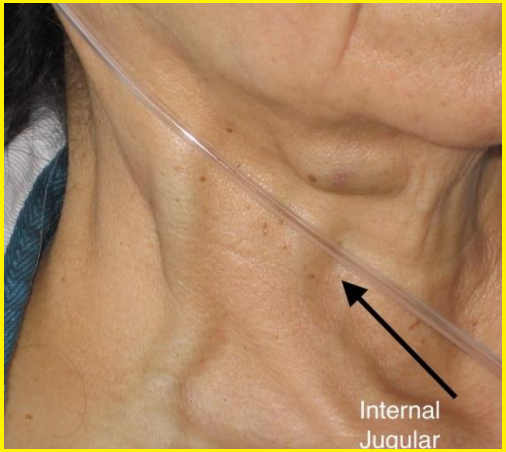
Ascites results from **transudation from hepatic and peritoneal veins**.

It is infrequently caused by

- Congestive heart failure. An increase in the preload or central venous pressure due to right ventricular dysfunction may cause direct liver damage due to increasing retrograde pressure to the venous and capillaries into the liver. Massive ascites in the setting of heart disease occurs most frequently in patients with tricuspid valve disease and constrictive pericarditis
- Nephrotic syndrome
- Pancreatitis

Pitting edema: Congestive heart failure



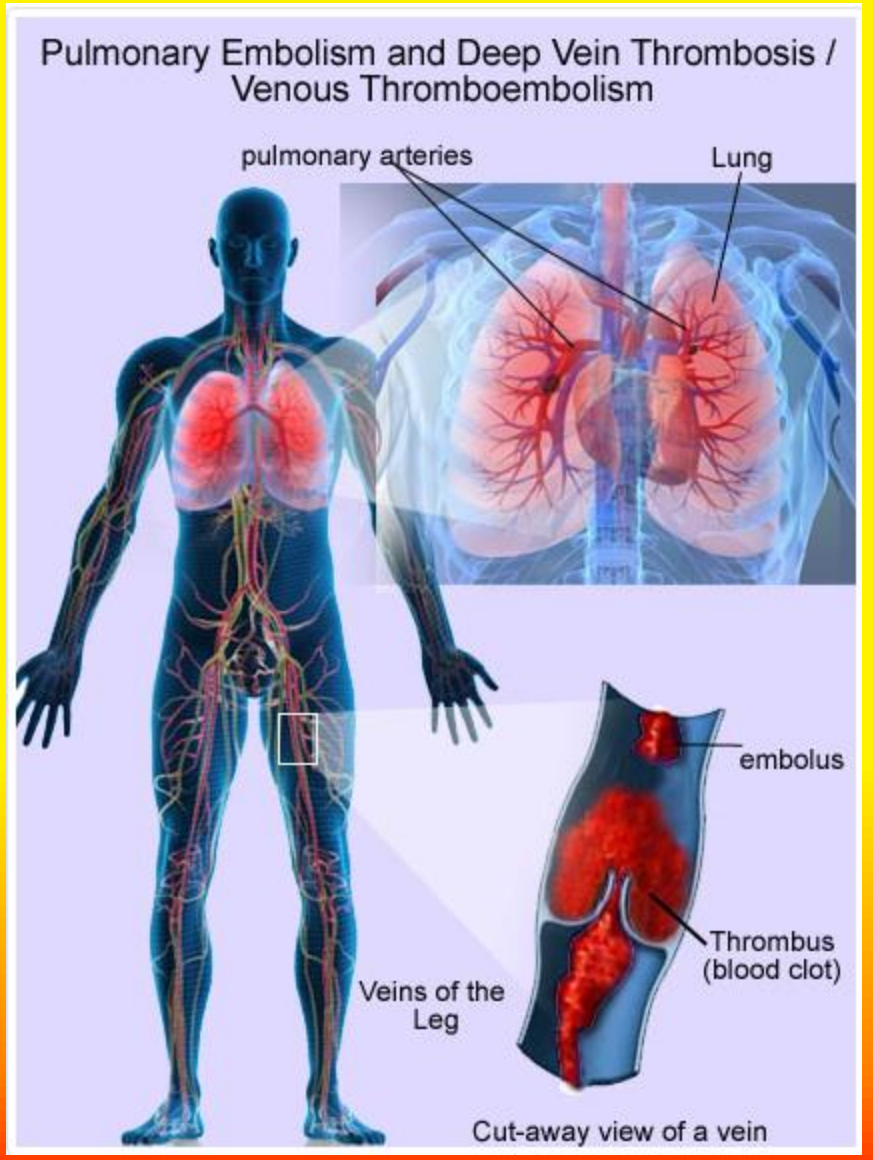


Pitting edema of the legs and ankle swelling

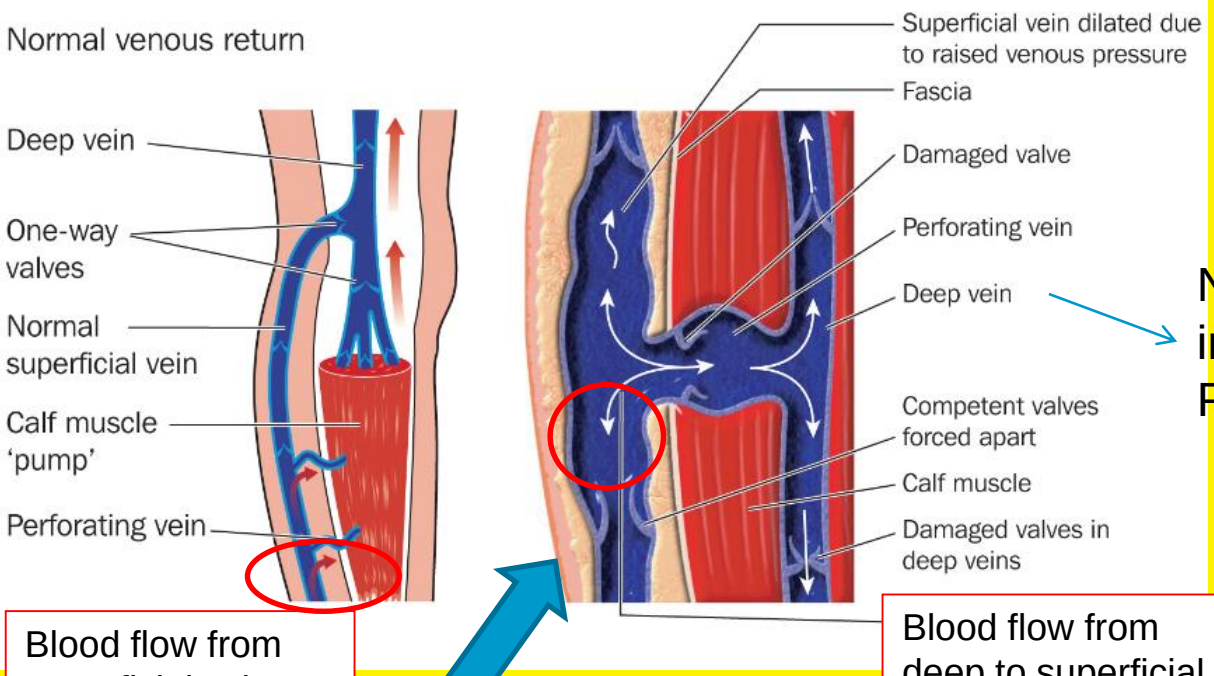


Ulcus crurus venosum vs deep vein thrombosis





Local edema: Chronic venous insufficiency



↑ risk for lung embolization

Note: thrombus formation in deep veins: Phlebothrombosis

↑risk

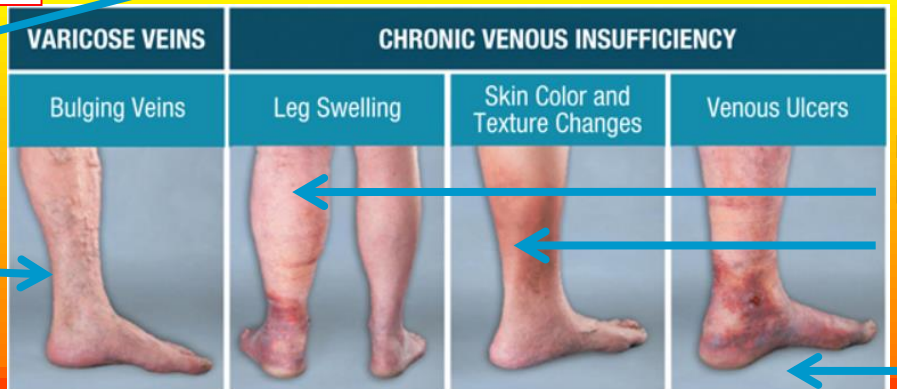
Chronic venous insufficiency

Blood flow from deep to superficial vein (back flow) mainly at calf muscle contraction

Varicose veins are superficial veins

Blood flow from superficial vein to deep vein (to heart) mainly at calf muscle contraction

Superficial thrombophlebitis



First edema due to ↑ hydrostatic pressure and later microangiopathy die to increased permeability

Pigmentation due to leaked erythrocytes

Ulcus crurus venosum due to Ischemia in the skin

1. Edema: classification based on cause

Increased hydrostatic pressure B and C

B: Kidney water and salt retention

- Acute or chronic renal failure with water and salt retention (nephritic syndrome)

C: Arteriolar dilatation

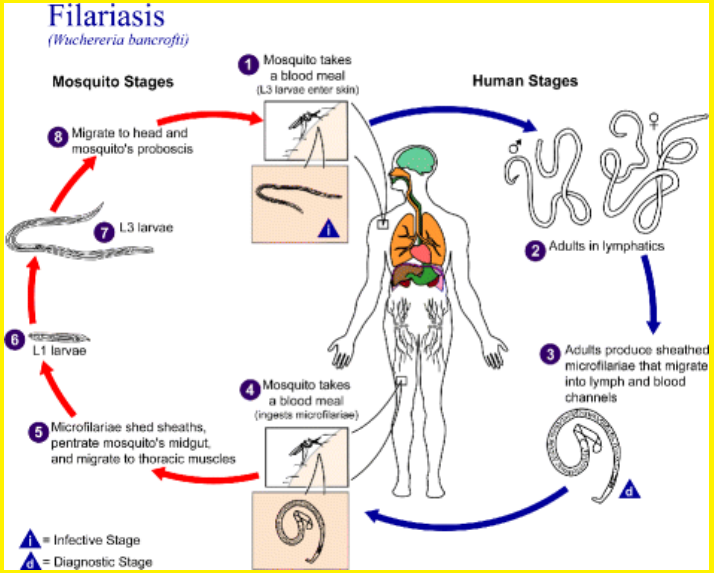
- Drugs, toxins, anaphylaxis, inflammation, sympathetic NS insufficiency or excessive heating or fever

! In arterial hypertension → constriction of arterioles→ capillary hydrostatic pressure NORMAL (except cerebral edema in hypertensive crisis)

1. Edema: classification based on cause: oncotic pressure, lymphatic edema and inflammation

- Decreased oncotic pressure
  - Nephrotic syndrome (>3.5g/day)
  - Hemorrhage (profuse bleeding)
  - Serious drainage from open wounds, burn,
  - Enteropathy
  - Malnutrition (Kwashiorkor)
  - Liver diseases (cirrhosis)
- Lymphatic edema: (local edema): tumor, infection or obstruction block
- Inflammatory edema: increased permeability of capillary membrane (exudate: more proteins in the inflammation)

Other classifications of edema



2. Classification based on extent
- Local/asymmetric edema: local edema do not activate hormones
- Generalized/systemic edema: usually extracellular e.g. anasarca

3. Classification based localization
- Intracellular: SIADH swollen red cell due to hyponatremia X shriveled cells
- Extracellular: ascites

4. Classification based on a role of kidney
- Primary edema: accumulation of Na^+ due to renal diseases „theory of overflow“
- Secondary: adaptive response (secondary activated RAAS) of kidney „theory of refilling“ e.g. The shift of fluid from the blood to the interstitium leading to secondary RAAS activation in ascites.

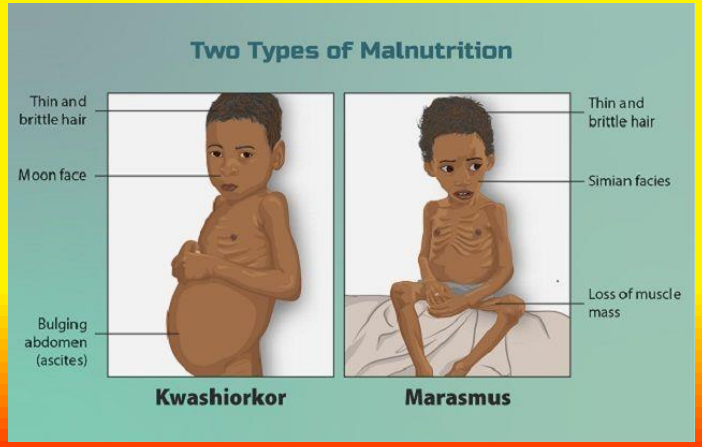
Other edemas

- Anasarca
- Myxedema
- Angioedema (Quincke’s edema)

Anasarca

a severe and generalized form of edema, with subcutaneous tissue swelling throughout the body and can involve the cavities of the body in addition to the tissues.
It reflects a severe disease state:

Liver failure
Kidney failure
Right-sided heart failure
Malignancy
Severe protein deficiency (decreased oncotic pressure)
Nephrotic syndrome
protein-losing enteropathies
Capillary leak syndrome



Myxedema

- a term used synonymously with severe hypothyroidism.
- refers to deposition of mucopolysaccharides in the dermis, which results in swelling of the affected area
- firm inelastic edema (opposite to pitting edema)

Angioedema (Quincke's edema)

- An allergic reaction: histamine
- An non-pitting and non-inflammatory edema of the lower layer of skin and tissue just under the skin or mucous membranes. The swelling may occur in the face (little supporting connective tissue), tongue, larynx (stridor), abdomen, or arms and legs. Often it is associated with hives (urticaria) and pruritus
- Other causes: an inherited C1-inhibitor protein deficiency (continuous activation of the complement system and kallikrein): bradykinin





Swollen eyelids, lips tongue

Infusion therapy (IV)

Crystalloids and colloids are plasma volume expanders used to increase a depleted circulating volume.

Crystalloid solutions are isotonic plasma volume expanders that contain electrolytes.

- sodium chloride 0.9% (normal saline 0.9%) (a side-effect of the excessive use: hyperchloraemic acidosis)
  - in isotonic extracellular dehydration
  - in sodium depletion
  - as vehicle or diluent of compatible drugs
- sodium lactate solutions (Ringer's lactate solution = Hartmann's solution)
  - in fluid resuscitation after blood loss
  - in metabolic acidosis: metabolic conversion of Ringer's lactate (no lactic acid in IV) to bicarbonate
- glucose solutions
  - in dehydration
  - in malnutrition to prevent ketosis,
  - carrier and diluent for parenteral nutrition (a side-effect of the excessive use: hyponatraemia or polyuria)
- additives such as potassium in hypokalaemia

Colloids are gelatinous solutions that maintain a high osmotic pressure in the blood. Colloids (=particles) stay in the intravascular spaces longer than crystalloids. side-effects can be hypertension, tachycardia, cardiac failure, pulmonary and peripheral oedema or an increased risk of anaphylaxis

- Albumin in cirrhosis and ascites,
- Dextran antithrombotic (antiplatelet), to reduce blood viscosity, and as a volume expander in hypovolaemia
- Dydroxyethyl starch (or hetastarch):

Cerebral edema: classification I and II

- Classification I
  - Local: local infection
  - Generalized: severe hyponatremia or hyponatremia fast onset
- Classification II
  - Intracellular: cytotoxic cerebral edema, some osmotic cerebral edema
  - Interstitial cerebral edema: hydrostatic, vasogenic, hydrocephalic, some osmotic cerebral edema

Cerebral edema: classification II

- A single mechanism or, more commonly, a combination of several mechanisms leads to brain edema.
  - Ionic (osmotic) cerebral edema
    - Main mechanism is the disruption of osmolarity: osmotically active particles
  - Vasogenic cerebral edema
    - Main mechanisms is a disruption of the blood-brain barrier
  - Cytotoxic cerebral edema
    - Any mechanism leading to low ATP
  - Interstitial (hydrocephalic) cerebral edema
    - Main mechanism is an obstruction to the outflow of cerebrospinal fluid
  - Hydrostatic cerebral edema
    - Main mechanism is ↑blood pressure
  - Other combined types of cerebral edema: e.g. High-altitude cerebral edema

Ionic (Osmotic) Cerebral edema

- The intact blood-brain.
  - Improper administration of intravenous fluids, isotonic or hypotonic.
  - Excessive water intake, SIADH, hyponatremis
  - Rapid correction of
    - ↑ Na^+ in hypernatremia
    - ↑ blood glucose in diabetic ketoacidosis or hyperosmolar hyperglycemic state.

Na^+ The cerebral edema during hypernatremia and hyponatremia I

Hypernatremia implies a plasma sodium level above
145 mEq/L (145 mmol/L) and characterized cellular dehydration.

- Acute hypernatremia: it has been present for 48 hours or less.
  Fast intervention or the exsiccation of the nerve cells
- Chronic hypernatremia: it has been present for longer than 48 hours.
  Slow intervention or brain edema

Hyponatremia represents a plasma sodium concentration below 135 mEq/L (135 mmol/L) and characterized cellular hyperhydration.

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- acute hyponatremia: it has developed over a period of less than 48 hours,
  - Fast intervention or fatal herniation (foramen magnum)
- chronic hyponatremia: it has been present for more than 48 hours
  - Slow intervention or osmotic demyelination syndrome

Na^+ Hypernatremia and hyponatremia II

BRAIN ADAPTATION TO SEVERE CHRONIC HYPERNATREMIA

- Endogenous/idiogenic osmoles (The brain can produce more particles from its cellular metabolism): taurine, glycine, glutamine, sorbitol and inositol and urea.
- Fast intervention: **cerebral edema**

BRAIN ADAPTATION TO HYPONATREMIA:

- increased intracranial pressure forces interstitial sodium and water out of the brain into the cerebrospinal fluid, ameliorating the increased brain volume. In addition, astrocytes begin to lose intracellular solutes, principally potassium and organic osmolytes.
- Fast intervention: **osmotic demyelination syndrome (ODS)**

Cerebral Edema during correction of hyperglycemic crises

1. On admission to hospital, profound dehydration and hypotension associated with a hyperglycaemic crisis will require stabilisation of blood pressure by rapid fluid infusion in preference to slow fluid administration as directed in guidelines. Thus, aggressive resuscitation begins with the administration of 1-1.5 L of saline in the first hour and should continue until blood pressure is stabilized (Step 1). It is imperative that insulin not be started during Step 1, because the addition of insulin during rapid fluid administration can cause unpredictable changes in blood glucose and serum osmolality.
2. Once the patient is adequately resuscitated, insulin should be started.....
    1. Insulin crosses the blood-brain barrier, but insulin is not required for glucose transport to most brain cells.
    2. Insulin lowers glycemia because insulin stimulates GLUT4-mediated glucose utilization by skeletal cells and adipose tissue. The drop in blood glucose is transmitted to the cerebral fluid with a delay (blood-brain barrier) and creates an osmotic gradient between the blood and the brain leading to interstitial cerebral edema.
        1. GLUT1 in glia and brain endothelial cells is <u>independent</u> of insulin
        2. GLUT3 in most neurons is <u>independent</u> of insulin
        3. GLUT4 in a few selected nuclei but common in adipose tissue, skeletal and cardiac cells is <u>sensitive</u> to insulin

Vasogenic cerebral edema

- Vasogenic: a disruption of the blood-brain barrier leading to the increased permeability of the capillary endothelium and accumulation of protein and fluid in the extracellular space.
    - The blood–brain barrier consists of astrocytes and pericytes joined together with adhesion proteins producing tight junctions

- This is the most common form of cerebral edema
- Examples:
    - Infections: meningitis
    - Inflammatory central nervous system: multiple sclerosis
    - Brain hemorrhage
    - Return of blood flow after an ischemic stroke (cytotoxic edema) can cause oxidative stress leading to dysfunction of the endothelial cells and disruption of the blood-brain barrier
    - Hypertensive encephalopathy

Cytotoxic Cerebral edema

- Cerebral ischemia
- Low ATP leading to low function of Na^+/K^+ pump, cellular retention of Na^+, Cl^- and water accumulation
- Examples:
    - intracerebral hemorrhage
    - ischemic stroke
    - acute liver failure: Ammonia accumulates in the blood stream and crosses the blood–brain barrier. Hyperammonemia in central nervous system (CNS) cells causes oxidative stress and mitochrondrial dysfunction, leading to damaged function of Na^+/K^+ pump and astrocytic cell swelling. Additionally, ammonia is converted to glutamine in CNS cells which acts as an osmolyte and draws further water into the cell through osmosis.

Cerebral edema in diabetic ketoacidosis (DKA)

- A complication of diabetes type I in children
- Ketoacidosis
  - A state of near absolute insulinopenia
  - A higher increase in counter regulatory hormones that support gluconeogenesis
  - Sever state of DKA can leads to combination of cytotoxic and vasogenic edema. Osmotic edema is less likely as the edema in DKA does not have to develop after fluid replacement like osmotic edema in pure hyperglycemia. On the other hand, hyperglycemia contributes to cellular injury by inducing dehydration a low perfusion

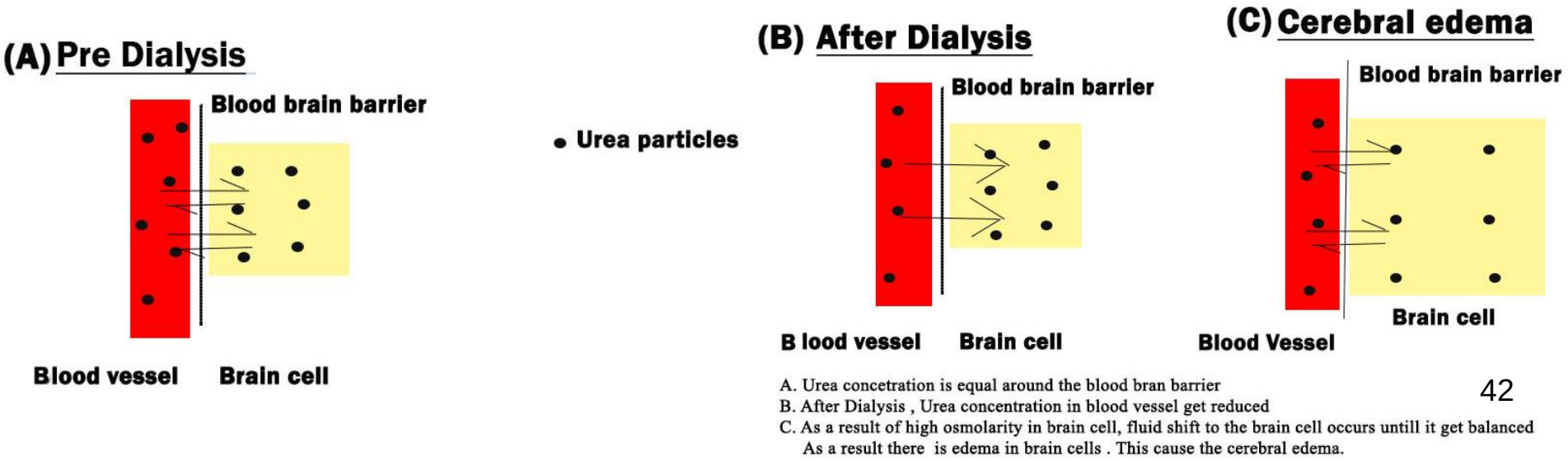
Interstitial (hydrocephalic) cerebral edema

- Hydrocephalus: an excess of fluid within the cerebral ventricles
- There is an obstruction to the outflow of cerebrospinal fluid within the ventricular system. The obstruction creates a rise in the intraventricular pressure and causes CSF to flow through the wall of the ventricles into the extracellular fluid within brain



The cerebral edema during the dialysis disequilibrium syndrome

It happens because the rapid fall in plasma urea concentration produced by hemodialysis in a uremic patient. In this setting, the removal of plasma urea occurs more quickly than urea can equilibrate across the hemo-cerebral barrier (delayed in the brain due to hemo-cerebral barrier), which promotes osmotic water movement into the brain. In the brain, the water shift can result in cerebral edema and acute neurologic dysfunction, changes that partially explain the dialysis disequilibrium syndrome





The cerebral edema during the dialysis disequilibrium syndrome

2 hypotheses:

- "Reverse Urea Effect"
  - The urea concentration in the CNS remains elevated because of its slower diffusion from the CNS to the blood than the diffusion of urea from the blood into the dialysate compartment.
  - Magnetic resonance imaging suggests **interstitial edema** and not intracellular (cytotoxic) edema: brain extracellular water increases and/or intracellular water decreases after hemodialysis

- "Idiogenic Osmoles" may contribute to "Reverse Urea Effect"
  - "Idiogenic Osmoles" hypothesis refers to idiogenic osmoles being responsible for cytotoxic edema.
  - Uremic toxins themselves could be the "idiogenic osmoles." These are questions that remain unanswered at the current time.

Hydrostatic cerebral edema

- An intact blood-brain
- Hypertensive encephalopathy occurs when pressure exceeds the upper limit of cerebral autoregulation, which normally maintains cerebral blood flow constant over a range of MAP from 60 to 150 mm Hg.
- Autoregulation:
  - ↑MAP leads to vasoconstriction in the brain
  - ↓MAP leads to vasodilatation in the brain
- Examples:
  - Severe arterial hypertension: about 1% of patients with hypertension will develop a hypertensive crisis (defined as a systolic pressure >180 mmHg or a diastolic pressure of > 110 mmHg).
  - Preeclampsia
  - Eclampsia is a seizure of CNS leading to hypoxic cerebral edema. It starts as pre-eclampsia and can be accompanied by tonic-clonic seizure with progression to coma. Pre-eclampsia presents with three main features: new onset of high blood pressure, protein in the urine and edema. Pathophysiology is not clear but there is decreased level of placental vasodilators compared to normal placenta.

Cerebral edema: manifestation

Intracranial Hypertension (ICH)

- defined as ICP greater than 20 mm Hg (normal range: 5–10 mm Hg)
    - defined by volume of brain. blood and cerebral fluid
    - Mean Arterial Pressure (MAP) - Intracranial Pressure (ICP) = Cerebral Perfusion Pressure (CPP)

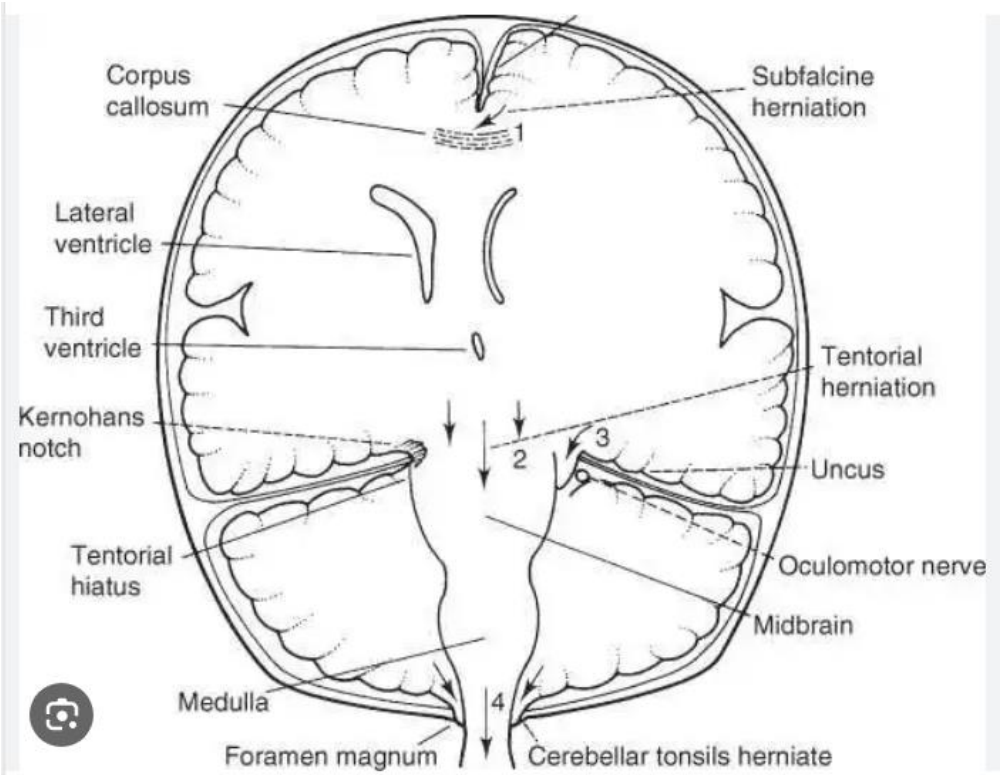
- poor effect of mean arterial pressure due to autoregulation:
- ↑MAP leads to vasoconstriction in the brain
- ↓MAP leads to vasodilatation in the brain

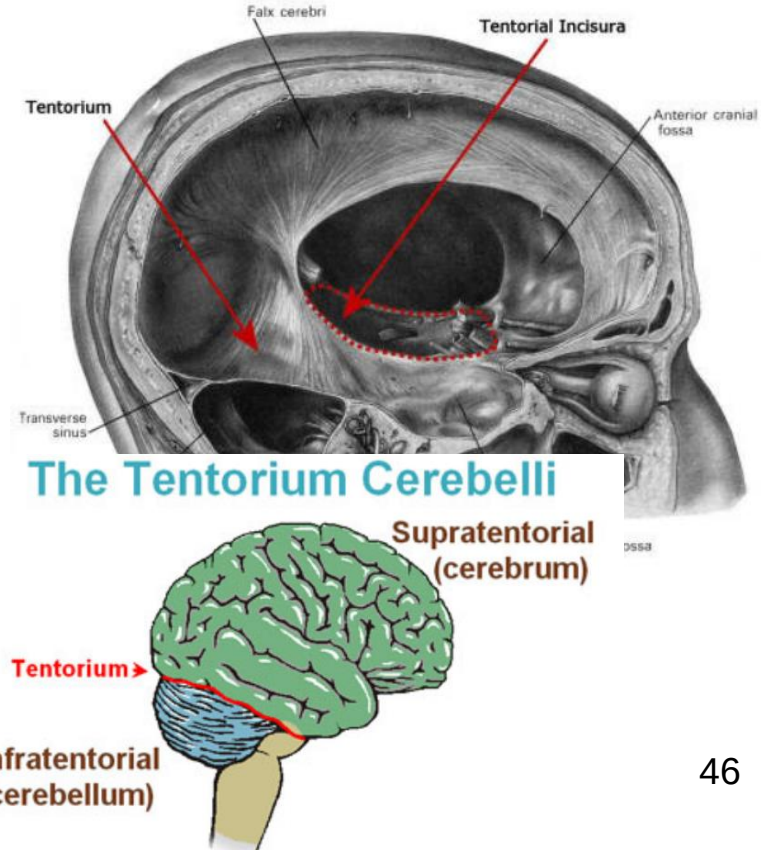
Stages:

- 100% compensation: due to ↓production and ↑resorption of cerebrospinal fluid (liquor)
- Decreased compensation: ↓ blood volume in the brain, unchanged brain perfusion due to ↑blood pressure
- Decompensation: ↑blood pressure but heart rate ↓, ischemia that ↑ CO_2 leading to vasodilatation that even more ↑ ICH, herniation

Herniation

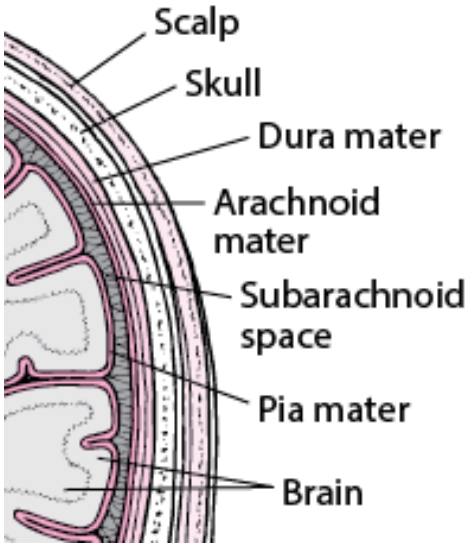
- Central tentorial herniation
  - In central herniation, the diencephalon and parts of the temporal lobes of both of the cerebral hemispheres are squeezed through a notch in the tentorium (dura mater) cerebelli (tentorial opening).
- Tonsillar herniation (downward cerebellar herniation)
  - The occipital conus is a herniation of the cerebellar tonsil through the foramen magnum. There is compression of the medulla oblongata.

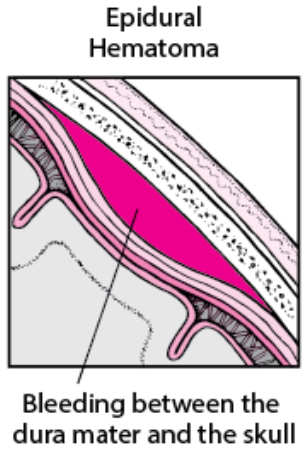


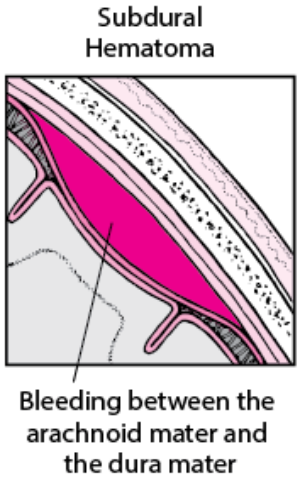


Cerebral edema and other causes of Intracranial Hypertension

- Cerebral edema (common):
  - Direct damage:↑ diffusion length
  - Indirect damage: compression
- Bleeding
  - Epidural (extradural) hematomas: an artery is the source of bleeding in 85%
  - Subdural hematomas: veins
  - Subarachnoid hematomas: rupture of aneurysms
  - Intracerebral hematomas: combination of hypertension (note: single no due to autoregulation MAP do not affect blood flow)) and arteriosclerosis
  - Intraventricular: bleeding into the brain's ventricular system
- Cancer
- Liquor circulation disruption







References

- Look up the particular theme in the following indexes
  - Porth's Pathophysiology: Concepts of Altered Health States (9th Edition) by Sheila Grossman, Carol Mattson Porth. Wolters Kluwer Health | Lippincott Williams & Wilkins, 2014.
  - McCance K. L., Huether S. E.: Pathophysiology. 8th Edition. Mosby, 2018.
  - Silbernagl S, Lang F. Color Atlas of Pathophysiology, 3rd Ed. Thieme, 2016.